

Class	Object Oriented Programming (OOP)	(two hours)
B2IT / B2ITE		Date : 7.5.2018

Name: _____

Reg. #: _____

10	8	8	6	4	14	

I. Choose the correct answers:

Q1. In a class definition, data or functions designated private are accessible

- (A) to any function in the program. (B) only to public members of the class.
(C) only if you know the password. ✓ (D) to member functions of that class.

Q2. A member function can always access the data

- (A) in any object of the class of which it is a member. (B) in the public part of its class.
(C) in the class of which it is a member. ✓ (D) in the object of which it is a member.

Q3. The dot operator (class member access operator) connects the two entities (left to right):

- ✓ (A) A class object and a member of that class (B) A class and a member of that class
(C) A class member and a class object (D) A class object and a class

Q4. Virtual functions allow you to

- (A) create an array of type pointer-to-base class that can hold pointers to derived classes.
✓ (B) use the same function call to execute member functions of objects from different classes.
(C) group objects of different classes so they can all be accessed by the same function code.
(D) create functions that can never be accessed.

Q5. Inheritance is a way to

- (A) pass arguments to objects of classes.
✓ (B) add features to existing classes without rewriting them.
(C) improve data hiding and encapsulation.
✓ (D) make general classes into more specific classes.

Q6. Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- (A) private members of Derv. ✓ (B) public members of Derv.
(C) protected members of Derv. (D) public members of Base.

Q7. A friend function can be used to

- ✓ (A) allow access to an unrelated class.
(B) allow access to classes whose source code is unavailable.
(C) mediate arguments between classes.

✓ (D) increase the versatility of an overloaded operator.

Q8. An abstract class is useful when

- (A) you want to defer the declaration of the class.
- ✓ (B) there are multiple paths from one derived class to another.
- (C) no objects should be instantiated from it.
- (D) no classes should be derived from it.

II. Fill in the blanks in each of the following:
(CHOOSE FIVE)

- Q9. Member functions defined inside a class definition are in line by default.
- Q10. The only technical difference between structures and classes in C++ is that the data and function are public in Structure & private in class by default.
- Q11. A constructor is executed automatically when an object is created.
- Q12. Deciding—after a program starts to execute—what function will be executed by a particular function call statement is called dynamic binding.
- Q13. In the UML, inheritance is called generalization.
- Q14. To be accessed from a member function of the derived class, data or functions in the base class must be public or protected.

III. State whether each of the following is true or false. If false, explain why:
(CHOOSE FOUR)

- Q15. In a class you can have more than one constructor with the same name.
True
- Q16. Data items in a class must be private.
False can be public or private
- Q17. If a base class contains a member function basefunc(), and a derived class does not contain a function with this name, can an object of the derived class access basefunc()?
True
- Q18. The user must always define the operation of the copy constructor.
False because the compiler provides default copy constructor
- Q19. A friend function can access a class's private data without being a member of the class.
True

IV. Direct questions:

Q20. What is the purpose of a class definition?

to describe how objects of classes will look when they are created

Q21. Explain the difference in operation between these two statements.

`person p1(p0);`

`person p1 = p0;`

There is no difference between them , because they are identical

Q22. If a base class and a derived class each include a member function with the same name, which member function will be called by an object of the derived class, assuming the scope-resolution operator is not used?

Driven class.

V. Write code of the following:

Q23. Assume that there is a class Derv that is derived from a base class Base . Write the declarator for a derived-class constructor that takes one argument and passes this argument along to the constructor in the base class.

`Derv( int h ) : Base ( h )`

Q24. Write the declaration for a pure virtual function called aragon that returns no value and takes no arguments.

`virtual void aragon ( ) = 0 ;`